

layout: post title: SwitchFree library prep date: '2024-03-29' categories: tags: [DNA, RNA, Library prep, Protocols]

projects: Kit test

SwitchFree library prep test

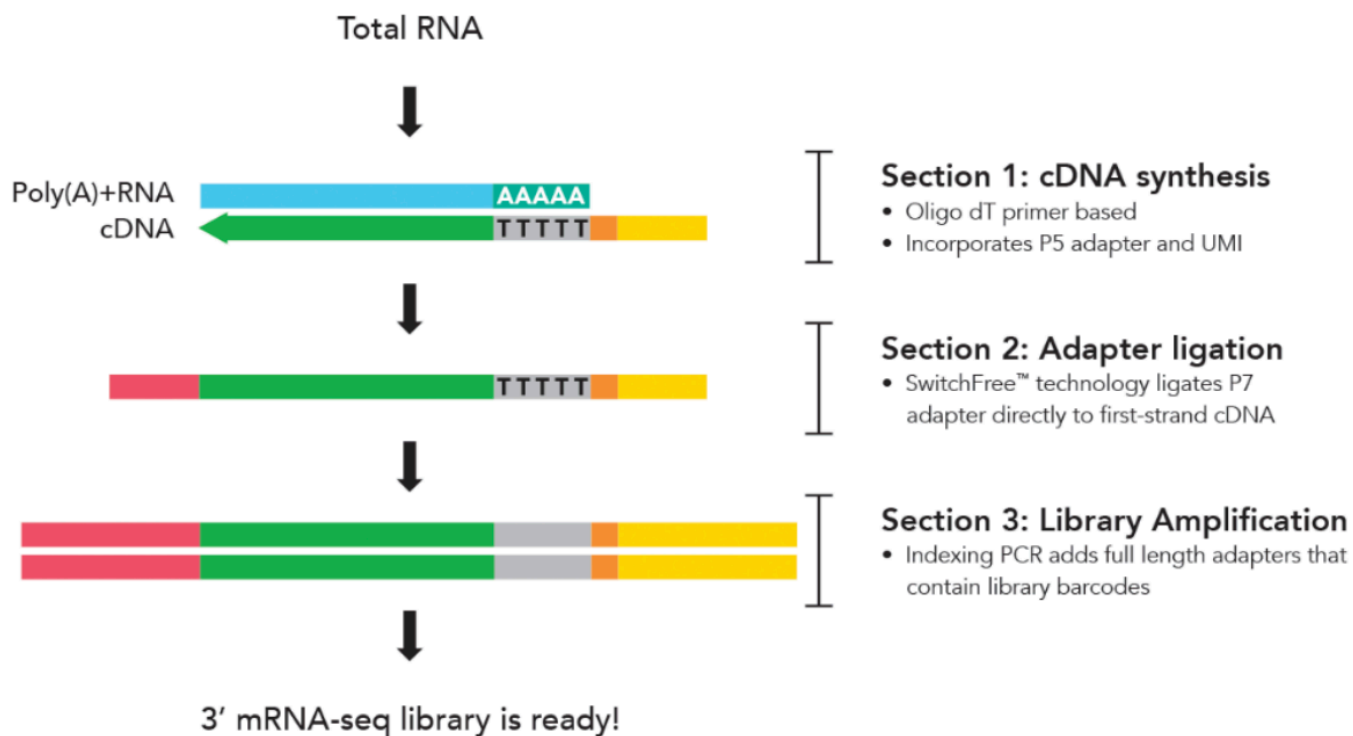
This post details the info about the SwitchFree library prep steps. Hollie got this kit as a test to see if we can produce adequate libraries for RNA sequencing with different species. I'm using the [Zymo-Seq SwitchFree 3' mRNA Library Kit](#). See Zymo's protocol [here](#).

We will be using samples 6, 7, 107, 108, 110, and 113 from the POC 2023 spawning project. These samples were extracted on [3/27/24](#). The kit needs a minimum of 10 ng of total RNA or a maximum of 500 ng of total RNA, which is a large range. Since some of the concentrations in these samples were low, we will be using 11 ng of total RNA as input. Here's a breakdown of input RNA volumes for each sample:

TubeID	TS RNA (ng/uL)	Strip tube #	RNA (uL)	Ultrapure water (uL)	Total starting volume (ul)	Primer
6	9.52	1	1.2	3.8	5.0	1
7	11.5	2	1.0	4.0	5.0	2
107	4.65	3	2.4	2.6	5.0	3
108	2.76	4	4.0	1.0	5.0	4
110	5.23	5	2.1	2.9	5.0	5
113	3.5	6	3.1	1.9	5.0	6

Here's the SwitchFree library prep workflow:

Zymo-Seq SwitchFree™ 3' mRNA Library Kit Workflow



Materials

- [Zymo-Seq SwitchFree 3' mRNA Library Kit](#)
- PCR tubes
- Thermocycler
- Mini centrifuge
- Aluminum beads (to keep things on ice)
- Magnetic stand for PCR tubes
- Kit contents

Zymo-Seq SwitchFree™ 3' mRNA Library Kit	R3008 (12 preps)	R3009 (96 preps)	Storage Temperature
PolyA R1 Reagent	60 µL	60 µL × 8	-80 °C
R2 Reagent	120 µL	120 µL × 8	-80 °C
Reaction Clean-up Solution	50 µL	50 µL × 8	-80 °C
3'mRNA L Reagent	180 µL	180 µL × 8	-80 °C
ZymoTaq™ PreMix	300 µL	300 µL × 8	-80 °C
Zymo-Seq™ UDI Primer Sets (Indexes 1-12) ¹	20 µL / Index	-	-80 °C
Zymo-Seq™ UDI Primer Plate (Indexes 1-96) ²	-	10 µL / Index	-80 °C
Select-a-Size MagBead Concentrate	300 µL	1.5 mL	4 °C
Select-a-Size MagBead Buffer	10 mL	50 mL	4 °C
Library Binding Solution	1 mL	10 mL	4 °C
DNA Wash Buffer (concentrate)	6 mL	48 mL	Room Temp.
DNA Elution Buffer	1 mL x 2	16 mL	Room Temp.
DNase/RNase-Free Water	1 mL × 2	10 mL	Room Temp.
Instruction Manual	1	1	-

Buffer preparation

Once buffers are prepared for a kit, they do not need to be prepared again.

- Add 300 µL of the Select-a-Size MagBead concentrate to 10 mL of Select-a-Size MagBead Buffer.
 - These should be prepared at least 5 days ahead of library prep.
- Add 24 mL of 100% ethanol to the DNA Wash Buffer concentrate. Store at room temperature.

Best practices

- Avoid multiple freeze thaws of all components, and aliquot components as necessary
- Remove enzymes from cold storage just before use and return to cold storage immediately after use
- Thaw and maintain all components on ice unless noted otherwise
- Flick to mix thawed components and briefly centrifuge prior to use
- After adding each component, mix by pipetting up and down 15-20 times. Briefly centrifuge after
- Pre-program thermal cycler with lid heating ON set to >100-105°C
- Turn on thermocycler day-of to preheat
- Pre-calculate how much to dilute RNA samples

Protocol

Section 1: cDNA Synthesis

Thermocycler settings:

Step	Temperature	Time
1	95°C	3 min
2	4°C	hold
3	45°C	20 min
4	42°C	hold
5	37°C	15 min
6	98°C	5 min
7	4°C	hold

- Thaw PolyA R1 Reagent and R2 Reagent on ice
- Prepare samples to correct volume in strip tubes
- Add 5 uL of PolyA R1 Reagent to each sample
- Mix by pipetting and centrifuge briefly
- Run Steps 1-2 in thermocycler
 - SKIP this step if RNA is degraded/low quality
- While in the thermocycler on the 4°C hold, add 10 uL of R2 Reagent to each sample
- Mix by pipetting
- Run Steps 3-4

- While Steps 3-4 are running, allow Reaction Clean-up Solution to equilibrate to room temperature
- While in the thermocycler on the 42°C hold, add 4 uL of Reaction Clean-up Solution to each sample
- Mix by pipetting and centrifuge briefly
- Immediately return to thermocycler
- Run Steps 5-7
- Remove tubes from thermocycler
- Add 26 uL of 95% ethanol to each sample
- Do bead clean-up with Select-a-Size MagBeads
 - Allow beads to equilibrate to room temperature >30 minutes before use
 - Resuspend beads immediately before use by shaking or vortexing until homogenous
 - Add 110 uL of Select-a-Size MagBeads to each sample
 - Pipette to mix until homogenous
 - Incubate for 5 minutes at room temperature
 - Put samples on magnetic stand for 5 mins or until beads have fully separated from solution
 - Aspirate slowly and discard supernatant
 - While the sample is still on the magnetic stand, add 200 uL of DNA Wash Buffer without disturbing the pellet
 - Aspirate slowly and discard supernatant
 - Repeat this wash step again
 - Keep the caps open to air-dry bead for 1 minute
 - Aspirate any residual wash buffer
 - Continue to air dry pellet until it appears matte without cracking (see picture below)



Figure 1. Over-dried beads are cracked and flakey, resembling dried mud. Under-dried beads are glossy and wet, like saturated mud. Optimally dried beads appear damp, but lack gloss.

- Remove sample from magnetic stand
- Add 10 μ L of DNA Elution Buffer to each sample
- Pipette to mix until homogenous
- Put samples on magnetic stand for 1-2 minutes or until elute is clear
- Move elute to new PCR tubes

Section 2: Adapter Ligation

Thermocycler settings:

Step 1: 95°C for 3 minutes, 4°C for 2 minutes, 4°C hold

Step 2: 25°C for 45 minutes (lid temperature at 40°C), 4°C hold

- Thaw 3'mRNA L Reagent on ice
- Put samples in thermocycler and run Step 1
- Remove samples from thermocycler and put on ice
- Add 15 uL of 3'mRNA L Reagent to each sample
- Mix by pipetting and centrifuge briefly
- Run Step 2
- Remove samples from thermocycler
- Add 25 uL of DNase/RNase free water to each sample
- Do bead clean-up with Select-a-Size MagBeads
 - Allow beads to equilibrate to room temperature >30 minutes before use
 - Resuspend beads immediately before use by shaking or vortexing until homogenous
 - Add 100 uL of Select-a-Size MagBeads to each sample
 - Pipette to mix until homogenous
 - Incubate for 5 minutes at room temperature
 - Put samples on magnetic stand for 5 mins or until beads have fully separated from solution
 - Aspirate slowly and discard supernatant
 - While the sample is still on the magnetic stand, add 200 uL of DNA Wash Buffer without disturbing the pellet
 - Aspirate slowly and discard supernatant
 - Repeat this wash step again
 - Keep the caps open to air-dry bead for 1 minute
 - Aspirate any residual wash buffer
 - Continue to air dry pellet until it appears matte without cracking (see picture above)
 - Remove sample from magnetic stand
 - Add 15 uL of DNA Elution Buffer to each sample
 - Pipette to mix until homogenous
 - Put samples on magnetic stand for 1-2 minutes or until elute is clear
 - Move elute to new PCR tubes

Section 3: Library amplification

Thermocycler settings:

Step	Temperature	Time	
1	95°C	10 min	
2	95°C	30 sec	} 500 ng = 13-14 cycles 100 ng = 15-17 cycles 10 ng = 19-21 cycles
3	58°C	30 sec	
4	72°C	60 sec	
5	--- Go to Step 2 ---		
6	72°C	7 min	
7	4°C	hold	

Given the low input for these preps and my previous experience with other Zymo kits, we are going to do 21 PCR amplification cycles.

- Thaw Index Primers and ZymoTaq Premix on ice
- Allow Library Binding Solution to equilibrate to room temperature >30 minutes before use
- Add 10 uL of unique Index Primer to each sample
- Mix by pipetting
- Add 25 uL of Zymo TaqPremix to each sample
- Mix by pipetting and centrifuge briefly
- Run thermocycler program above
- Remove samples from thermocycler
- Add 50 uL of DNase/RNase free water to each sample
- Add 80 uL of Select-a-Size MagBeads to each sample
- Pipette until homogenous
- Incubate for 5 minutes at room temperature
- Place samples on magnetic stand for 5 minutes or until beads have separated from solution
- Aspirate slowly and discard supernatant
- Remove samples from magnetic stand
- Add 100 uL of DNA Elution Buffer to beads
- Pipette until homogenous
- Add 80 uL of Library Binding Solution to each sample
- Pipette until homogenous
- Incubate for 5 minutes at room temperature
- Do bead clean-up with Select-a-Size MagBeads (beads already added to samples)
 - Put samples on magnetic stand for 5 mins or until beads have fully separated from solution
 - Aspirate slowly and discard supernatant
 - While the sample is still on the magnetic stand, add 200 uL of DNA Wash Buffer without disturbing the pellet
 - Aspirate slowly and discard supernatant

- Repeat this wash step again
- Keep the caps open to air-dry bead for 1 minute
- Aspirate any residual wash buffer
- Continue to air dry pellet until it appears matte without cracking (see picture above)
- Remove sample from magnetic stand
- Add 15-20 uL of DNA Elution Buffer to each sample
- Pipette to mix until homogenous
- Put samples on magnetic stand for 1-2 minutes or until elute is clear
- Move elute to new PCR tubes

THIS IS THE FINAL 3' mRNA-seq LIBRARY. STORE AT -20°C.

QC

Run [DNA Tapestation](#) to visualize libraries. Here's an example of what the library should look like on a Tapestation:

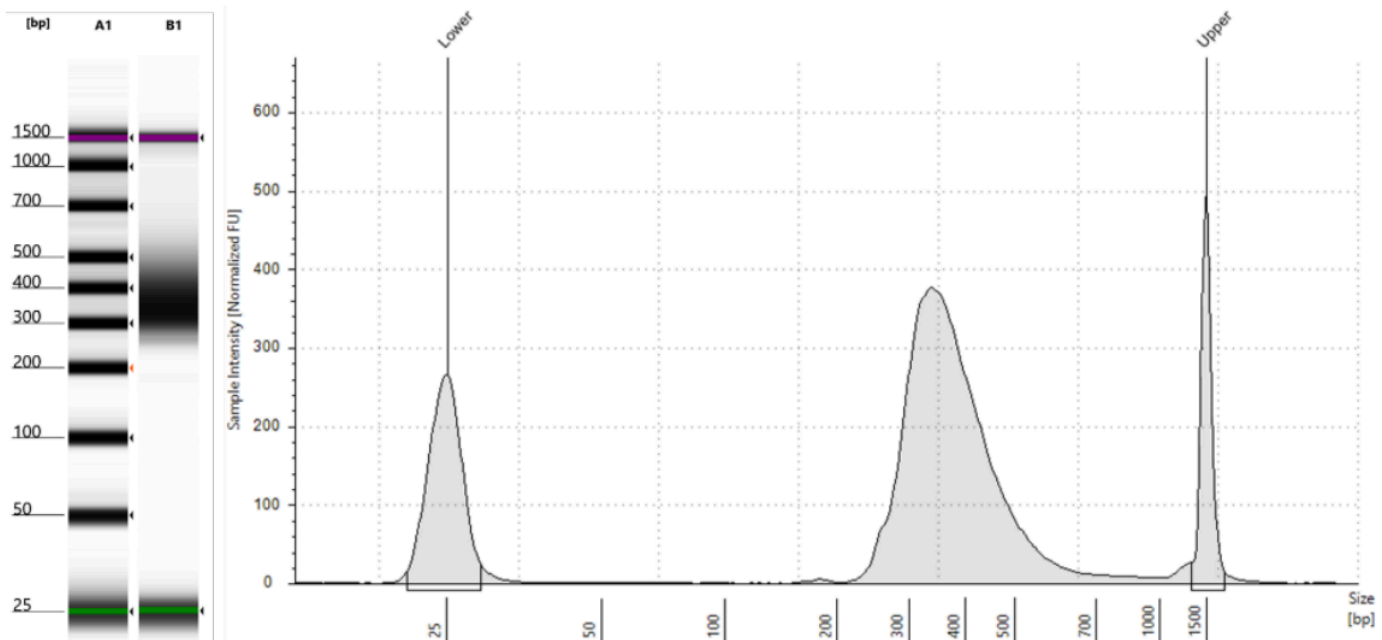
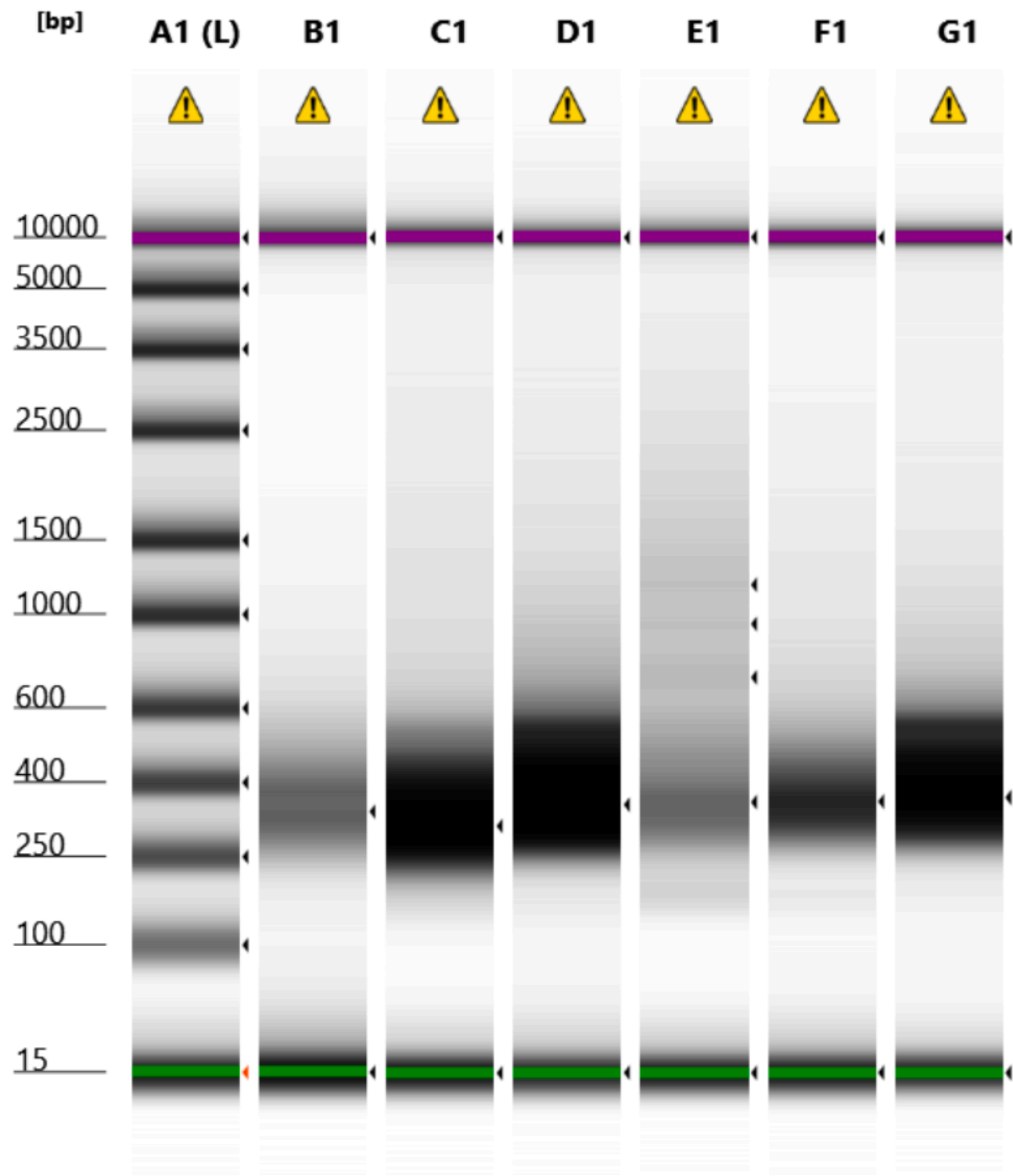


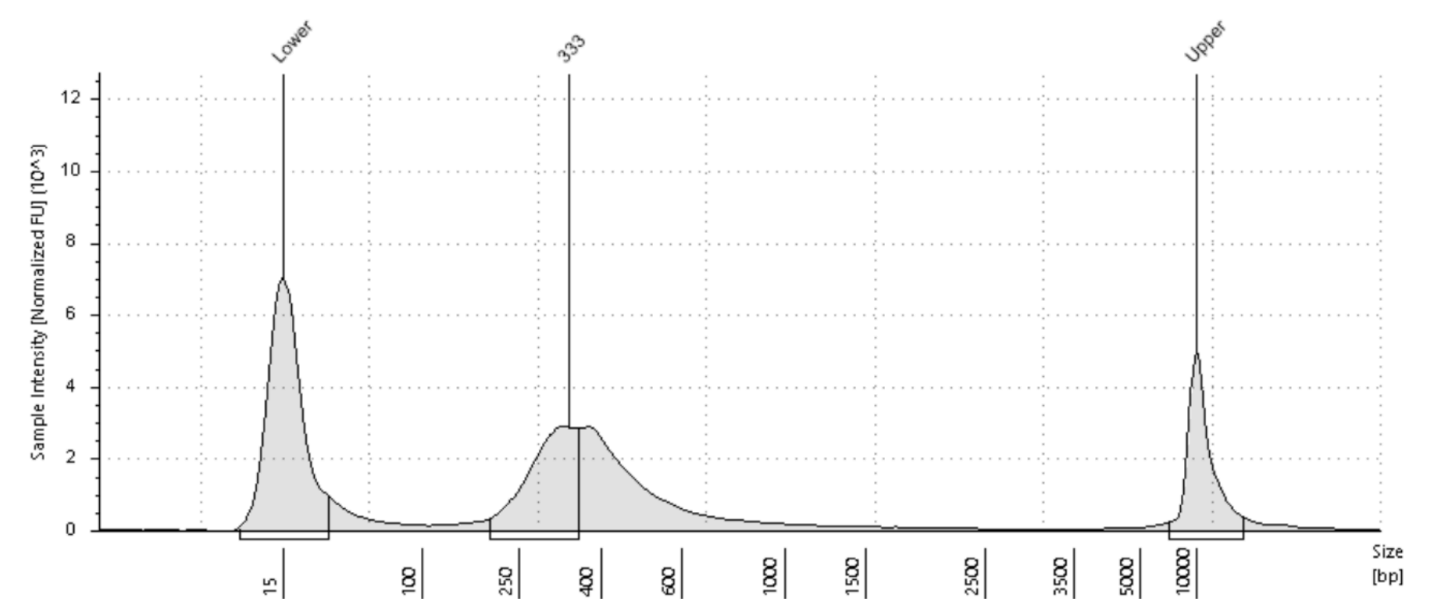
Figure 1. Agilent 4150 TapeStation® HSD1000 electropherogram of a typical Zymo-Seq SwitchFree™ 3' mRNA Library using 10 ng of HeLa total RNA (RIN > 9) and indexed using 19 PCR cycles. Fragment sizes mainly range from 250 to 700 bp. The peaks at 25 and 1500 bp are the interior standards of the High Sensitivity D1000 tape.

I ran the tapestation on the libraries on 3/31/24. See the full report [here](#).

Filename: 2024-03-31 - 14.05.45.D5000



B1: 6



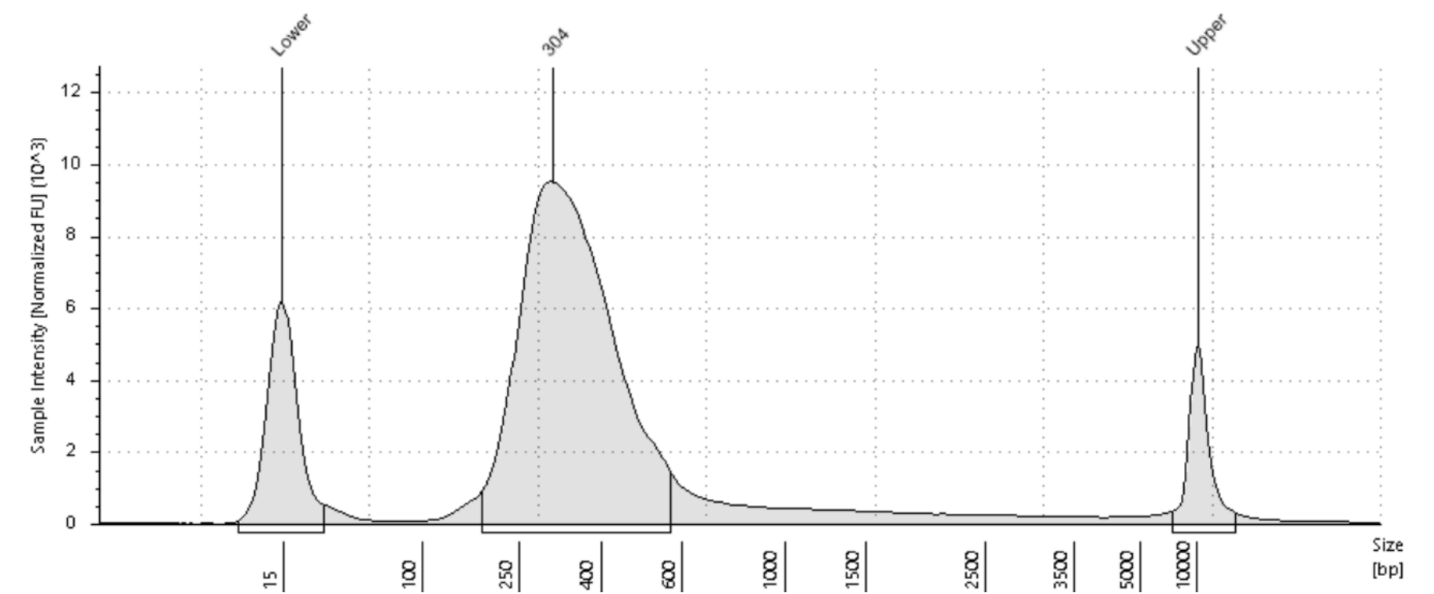
Sample Table

Well	Conc. [ng/μl]	Sample Description	Alert	Observations
B1	4.97	6		Caution! Expired ScreenTape device

Peak Table

Size [bp]	Calibrated Conc. [ng/μl]	Assigned Conc. [ng/μl]	Peak Molarity [nmol/l]	% Integrated Area	Peak Comment	Observations
15	7.10	-	728	-		Lower Marker
333	4.97	-	23.0	100.00		
10000	3.25	3.25	0.500	-		Upper Marker

C1:7



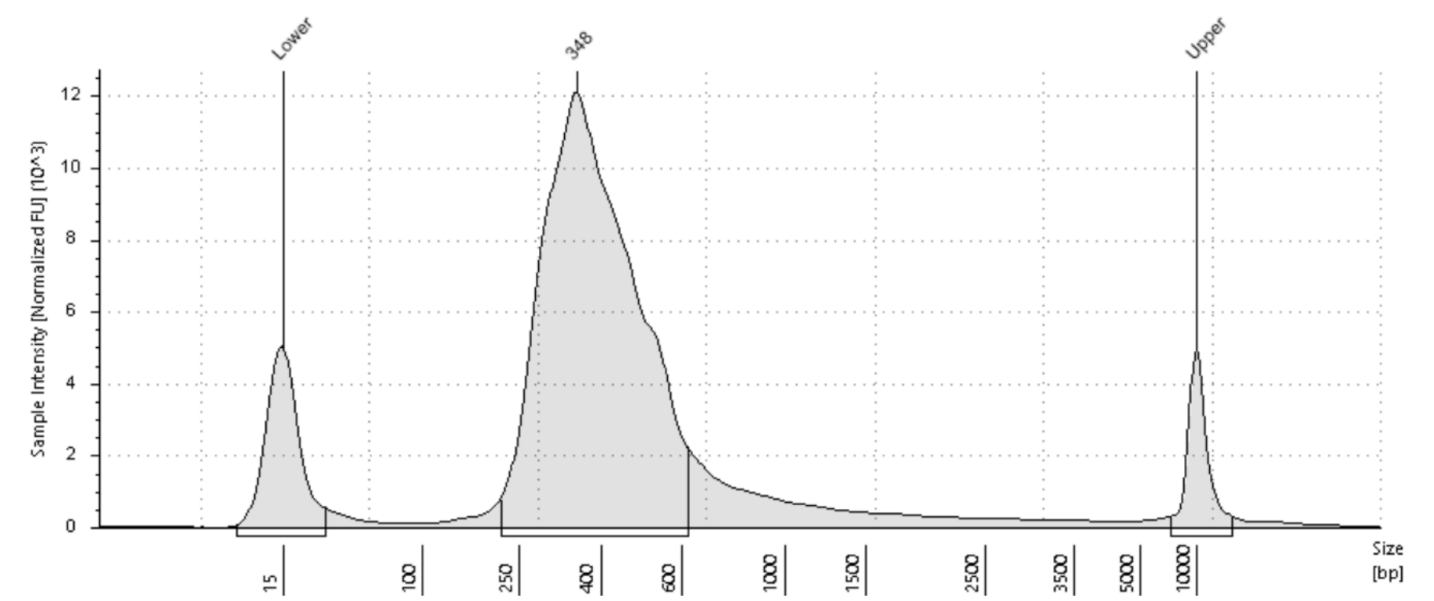
Sample Table

Well	Conc. [ng/μl]	Sample Description	Alert	Observations
C1	36.8	7		Caution! Expired ScreenTape device

Peak Table

Size [bp]	Calibrated Conc. [ng/μl]	Assigned Conc. [ng/μl]	Peak Molarity [nmol/l]	% Integrated Area	Peak Comment	Observations
15	6.90	-	708	-		Lower Marker
304	36.8	-	186	100.00		
10000	3.25	3.25	0.500	-		Upper Marker

D1: 107



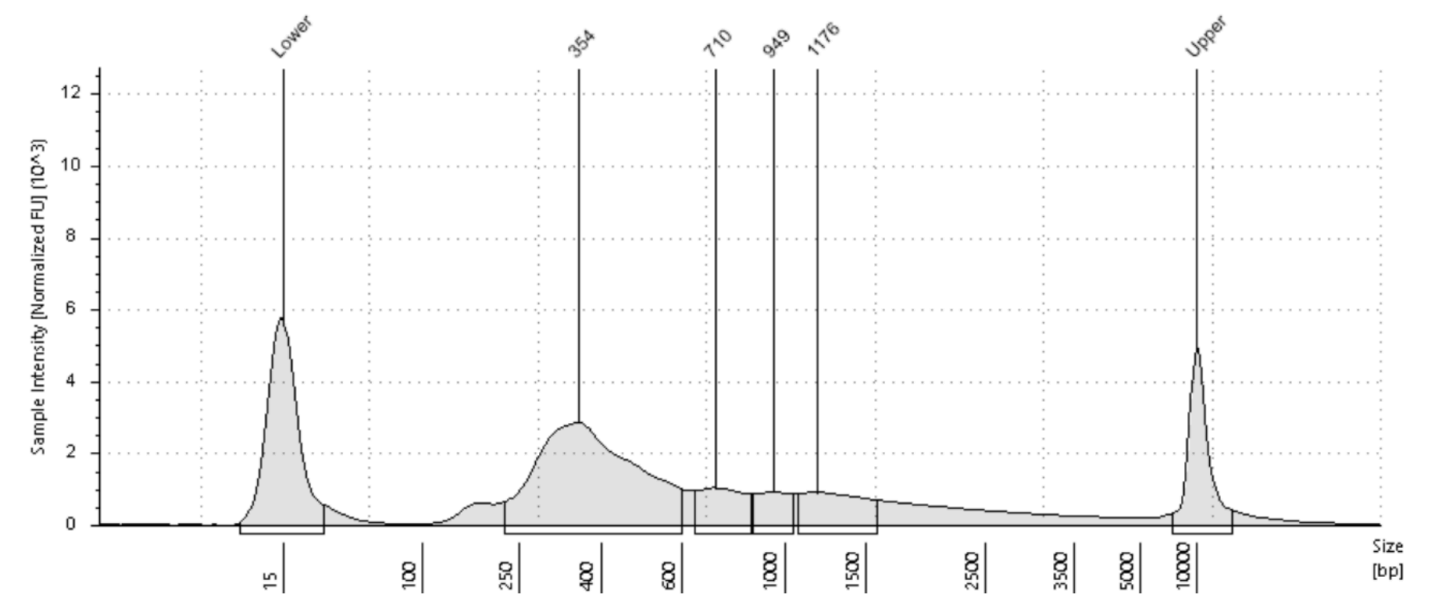
Sample Table

Well	Conc. [ng/ul]	Sample Description	Alert	Observations
D1	47.8	107		Caution! Expired ScreenTape device

Peak Table

Size [bp]	Calibrated Conc. [ng/ul]	Assigned Conc. [ng/ul]	Peak Molarity [nmol/l]	% Integrated Area	Peak Comment	Observations
15	6.05	-	621	-		Lower Marker
348	47.8	-	212	100.00		
10000	3.25	3.25	0.500	-		Upper Marker

E1: 108



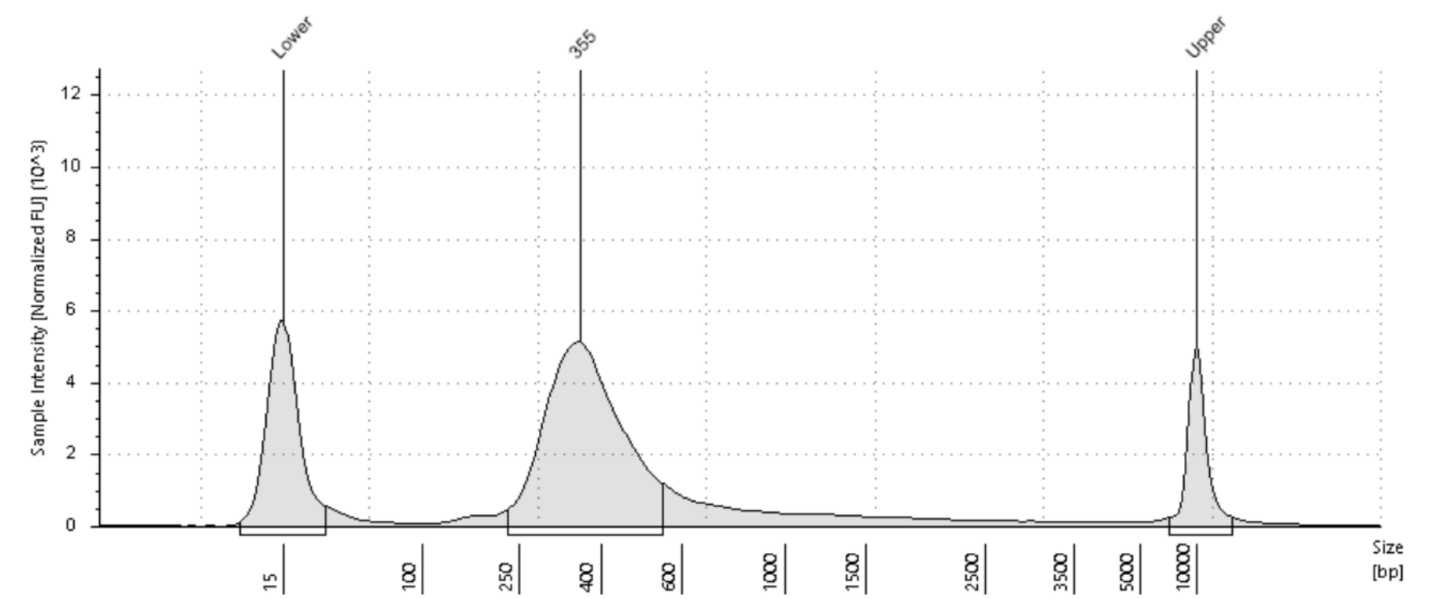
Sample Table

Well	Conc. [ng/ul]	Sample Description	Alert	Observations
E1	18.4	108		Caution! Expired ScreenTape device

Peak Table

Size [bp]	Calibrated Conc. [ng/ul]	Assigned Conc. [ng/ul]	Peak Molarity [nmol/l]	% Integrated Area	Peak Comment	Observations
15	6.69	-	686	-		Lower Marker
354	12.3	-	53.6	66.95		
710	2.11	-	4.58	11.47		
949	1.43	-	2.32	7.76		
1176	2.54	-	3.33	13.82		
10000	3.25	3.25	0.500	-		Upper Marker

F1: 110



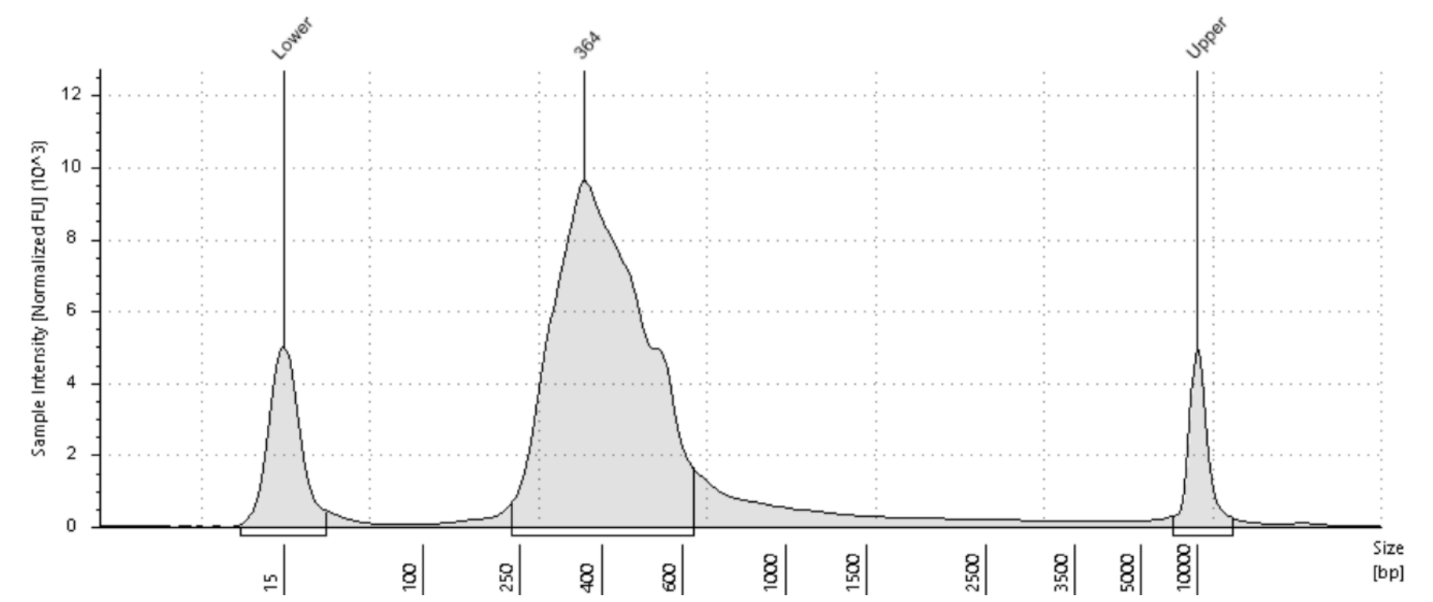
Sample Table

Well	Conc. [ng/μl]	Sample Description	Alert	Observations
F1	16.6	110		Caution! Expired ScreenTape device

Peak Table

Size [bp]	Calibrated Conc. [ng/μl]	Assigned Conc. [ng/μl]	Peak Molarity [nmol/l]	% Integrated Area	Peak Comment	Observations
15	6.65	-	682	-		Lower Marker
355	16.6	-	71.8	100.00		
10000	3.25	3.25	0.500	-		Upper Marker

G1: 113



Sample Table

Well	Conc. [ng/ul]	Sample Description	Alert	Observations
G1	37.3	113		Caution! Expired ScreenTape device

Peak Table

Size [bp]	Calibrated Conc. [ng/ul]	Assigned Conc. [ng/ul]	Peak Molarity [nmol/l]	% Integrated Area	Peak Comment	Observations
15	5.87	-	602	-		Lower Marker
364	37.3	-	157	100.00		
10000	3.25	3.25	0.500	-		Upper Marker

Tapestation looks great!! Sample 108 looks a little wonky, perhaps due to a PCR bubble product (read more [here](#)). Libraries with a bubble product can still be sequenced, which is good. Need to discuss next steps with Hollie + Ariana.